

Before the Levy: Announcement Pass-Through and Carbon-Price Calibration in Denmark’s Livestock Tax

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Abstract

Denmark announced in June 2024 that it would introduce a greenhouse-gas tax on livestock emissions. I estimate the anticipatory response of beef prices using scraped product–store prices and the official Danish consumer price index (CPI). The respective difference-in-differences (DiD) estimates are 4.4 and 6.4 log points; aggregate synthetic DiD is positive but imprecise. A model of planned supply with costly intertemporal adjustment shows that a credible future emissions tax reduces output both at implementation and beforehand, raising the pre-implementation price. Staggered contract renewal maps an immediate shadow-price response into gradual retail pass-through. By contrast, a fixed-stock slaughter model predicts more current supply and a lower current price. The positive estimates therefore favor planned contraction over pure pre-emptive slaughter, although prices alone do not identify quantities or emissions. I also pool ten comparable social-cost-of-carbon (SCC) estimates. The equal-paper-weight mean is USD 158/tCO₂ in 2024 dollars, with a hierarchical 95 percent interval of USD 77–363. Accounting comparisons under both the DKK 120/tCO₂ average output burden and the DKK 300/tCO₂ marginal abatement incentive remain sensitivity exercises, not estimates of incidence or welfare.

1 Introduction

Pricing agricultural greenhouse-gas emissions is difficult because the emissions are biologically heterogeneous, farm-level measurement is costly, and food prices are politically salient. A further complication is timing. If a future tax is credible, firms may revise contracts, procurement, and retail prices when the policy is announced rather than when the first liability is paid. Estimates that begin only at implementation can therefore miss part of the policy response.

On 24 June 2024, the Danish government and the parties to the Green Tripartite announced an agreement to introduce a greenhouse-gas tax on livestock emissions (Danish Government, 2024). The measure is an upstream agricultural emissions tax rather than a retail meat excise, and implementation was scheduled for later years. This paper asks whether the announcement changed supermarket beef prices and how the estimated response compares with the price signal implied by the broader SCC literature.

The empirical contribution is to triangulate the same research question with two sources whose weaknesses are complementary. The scraped data provide a large, timely product–store panel but are non-official and require transparent text classification and unit normalization. Statistics Denmark’s CPI is official and consistently classified, but supplies only a small cross-section of product indices. The microdata DiD estimate is 0.0437 (standard error 0.0163); the aggregate lags-only DiD estimate is 0.0645 (HAC standard error 0.0306). Their difference is not statistically meaningful. Synthetic DiD provides a complementary aggregate robustness check and gives 0.0760, although its placebo interval is wide.

The second contribution is a literature-based calibration that does not privilege a single SCC study. I harmonize one preferred global total-SCC estimate per paper to 2024 dollars. Estimates reported for years other than 2030 enter when the paper does not imply a time path, in which case the estimate is treated as time-invariant. Published uncertainty ranges are retained and propagated through a hierarchical paper bootstrap. This produces a ten-paper mean of USD 158/tCO₂ and a wide interval of USD 77–363. The exercise is a transparent synthesis of heterogeneous studies, not a claim that their model ranges are conventional sampling confidence intervals.

The broader contribution concerns announcements as policy instruments. Cattle production has biological lags and costly adjustment (Rosen et al., 1994). Environmental-economics models and evidence show that agents anticipate future regulation (Smulders et al., 2012; Rittenhouse and Zaragoza-Watkins, 2018), while emissions costs can pass through into product prices (Fabra and Reguant, 2014). Building on these ideas, I develop a transparent planned-supply model in which the announcement creates an expected future output wedge. When current and future production plans are linked, credibility makes contraction begin before collection. This is a commodity-market application of the general expectations channel highlighted by Mario Draghi’s *whatever it takes* commitment (Draghi, 2012), not an assertion that monetary and environmental policy operate identically.

The paper focuses exclusively on beef. The estimand is especially relevant because cattle are the most emissions-intensive major livestock product and prominent in the Danish agreement. Restricting attention to a pre-specified commodity also avoids selecting among a large family of weakly identified commodity-specific estimates. Section 2 develops the model. Sections 3–5 describe the data, estimators, and results. Section 6 presents the SCC accounting exercise, and Sections 7 and 8 discuss interpretation and limitations.

2 Theoretical Framework

2.1 A credible future tax and planned supply

There are two decision dates: the announcement date, normalized to 0, and the implementation date $T > 0$. Let $Q_t \geq 0$ denote aggregate beef output at date $t \in \{0, T\}$, let $P_t(Q_t)$ be inverse demand with $P'_t(Q_t) < 0$, and let $C_t(Q_t)$ be the industry’s private production cost with $C'_t(Q_t) > 0$

and $C_t''(Q_t) \geq 0$. Define gross private surplus net of production cost as

$$G_t(Q) \equiv \int_0^Q P_t(x) dx - C_t(Q). \quad (1)$$

Thus $G_t'(Q) = P_t(Q) - C_t'(Q)$ and $G_t''(Q) = P_t'(Q) - C_t''(Q) < 0$. One unit of output generates $\chi > 0$ tons of CO2-equivalent emissions. The statutory tax at T is τ_T DKK per ton of CO2e. Under risk neutrality, the announcement creates the expected output wedge

$$\omega \equiv \chi E_0[\tau_T], \quad E_0[\tau_T] = \pi \tau \quad (2)$$

in the special case in which the tax equals $\tau > 0$ with implementation probability $\pi \in [0, 1]$ and zero otherwise. Here $E_0[\cdot]$ is the expectation conditional on information at announcement.

Production planning is intertemporal. Changing planned output between dates costs $\kappa(Q_T - Q_0)^2/2$, where $\kappa \geq 0$ summarizes biological lags, capacity adjustment, and procurement commitments. Let $\beta \in (0, 1]$ be the one-period discount factor and $\delta \equiv \beta^T$ the discount factor from announcement to implementation. A competitive equilibrium can be represented as the unique maximizer of the strictly concave potential

$$G_0(Q_0) + \delta \left[G_T(Q_T) - \omega Q_T - \frac{\kappa}{2} (Q_T - Q_0)^2 \right]. \quad (3)$$

This representation does not make firms internalize consumer surplus: its first-order conditions reproduce market clearing and price-taking supply in a competitive industry. It also abstracts from literal herd accounting; Q_t is planned market output and κ is a reduced-form link between production plans.

Proposition 1 (Anticipatory contraction) *Suppose the solution to (3) is interior and $\kappa > 0$. A larger expected implementation wedge reduces planned output both before and at implementation and raises the announcement-period market price:*

$$\frac{\partial Q_0}{\partial \omega} < 0, \quad \frac{\partial Q_T}{\partial \omega} < 0, \quad \frac{\partial P_0}{\partial \omega} > 0. \quad (4)$$

The pre-implementation response vanishes when the policy is not credible ($\pi = 0$), adjustment is costless ($\kappa = 0$), or implementation is arbitrarily distant ($\delta \rightarrow 0$).

The proof is in Appendix A. For a closed form, let demand and costs be stationary and linear-quadratic: $P(Q) = a - bQ$ and $C(Q) = cQ + \gamma Q^2/2$, where $a > c$, $b > 0$, and $\gamma \geq 0$. Write $h \equiv b + \gamma$

and $Q^* \equiv (a - c)/h$. Then

$$Q_0 = Q^* - \frac{\delta\kappa}{h[h + \kappa(1 + \delta)]}\omega, \quad (5)$$

$$Q_T = Q^* - \frac{h + \delta\kappa}{h[h + \kappa(1 + \delta)]}\omega, \quad (6)$$

$$\Delta P_0 = \frac{b\delta\kappa}{h[h + \kappa(1 + \delta)]}\pi\tau\chi. \quad (7)$$

Equation (7) makes the mechanism explicit. The price response is larger when implementation is more credible (π), the announced rate (τ) or emissions intensity (χ) is higher, implementation is closer (larger δ), or intertemporal adjustment is more costly (larger κ). The last two claims follow because the response coefficient is increasing in both δ and κ ; it remains bounded and is not a claim that the observed retail-price change equals the future tax wedge.

2.2 Observed retail prices and contract renewal

The model price $p_t^* \equiv P_t(Q_t)$ is the flexible wholesale or shadow price. Retail pass-through can be delayed by contracts. Let $L_i \geq 0$ be contract i 's first renewal date after announcement, $F_L(t) = \Pr(L_i \leq t)$ its cumulative distribution, and $\lambda_i \in [0, 1]$ its pass-through share. If λ_i is independent of L_i with mean λ , an immediate flexible-price change Δp_t^* is observed as

$$\Delta p_t^R = E[\lambda_i \mathbf{1}\{L_i \leq t\}] \Delta p_t^* = \lambda F_L(t) \Delta p_t^*, \quad (8)$$

where p_t^R is the retail price and $\mathbf{1}\{\cdot\}$ is an indicator. Staggered renewal therefore produces gradual measured pass-through even when expectations move at announcement. Emergence around official event month 6 and scraped event month 8 is consistent with semiannual or dispersed contracting, but does not identify a contract duration.

2.3 Environmental benchmark and a sign-reversing boundary case

Let environmental damages at date t be $D_t(E_t)$, where emissions are $E_t = \chi Q_t$, $D'_t(E_t) > 0$, and D_t is measured in DKK. If $SCC_t \equiv D'_t(E_t)$ is marginal damage per ton of CO₂e, the standard Pigouvian benchmark is

$$\tau_t^P = SCC_t. \quad (9)$$

Proposition 2 (Local welfare effect) *Suppose adjustment costs are real resource costs and the assumptions of Proposition 1 hold. Starting from no emissions price, a small increase in a credible future tax raises welfare whenever marginal environmental damage is positive. More generally, an increase in ω raises welfare locally if the expected output wedge does not exceed marginal implementation-date damage per unit of output, $\omega \leq \chi D'_T(E_T)$; the induced pre-implementation contraction supplies an additional benefit when $D'_0(E_0) > 0$.*

This result establishes a local corrective role for a credible announcement; it does not show that the Danish rate is globally optimal or identify welfare from retail prices. The relevant calibration margin also matters. The announced 2030 schedule has a DKK 300/tCO₂e marginal abatement incentive, while the 60 percent per-animal deduction creates an effective average output burden of DKK 120/tCO₂e (Danish Government, 2024; Danish Ministry of Taxation, 2024).

Finally, the sign in Proposition 1 is not mechanical. Suppose instead that a fixed herd stock H must be slaughtered across the two dates, so $Q_0 + Q_T = H$, and there is no margin for planned herd contraction. A future output tax makes later slaughter relatively less attractive, shifts slaughter to date 0, and therefore gives $\partial Q_0 / \partial \omega > 0$ and $\partial P_0 / \partial \omega < 0$. This preemptive-slaughter channel is the livestock analogue of an anticipation or green-paradox response (Smulders et al., 2012). Positive observed prices favor planned contraction over this pure fixed-stock mechanism, but price-only evidence cannot establish that aggregate quantities or emissions fell.

3 Data and Panel Construction

3.1 Scraped grocery-price microdata

The first source is the Danish grocery-price history assembled by dagligepriiser.dk. A raw record identifies a store, product, product description, date, price, and—when available—package quantity and unit. I retain observations with a valid positive price and date. Repeated observations within a product–store–month are averaged, so the unit of observation for the main panel is a product–store pair in a calendar month.

Prices are converted to DKK per kilogram for mass units and DKK per litre for volume units. Reported package quantities take precedence. When quantity or unit fields are absent, the procedure searches the product description for a number followed by kg, g, litre, ml, or cl. Grams, millilitres, and centilitres are converted to their base units before dividing price by quantity. Items measured only per piece and observations with missing, non-positive, or unsupported quantities are excluded. Because normalized prices remain right-skewed, estimation uses their natural logarithm and Table 1 reports medians and quartiles as well as means.

Product classification is deterministic and auditable. Names and source categories are case-folded; Danish characters and common encoding variants are transliterated; and short tokens are matched at word boundaries. Beef is assigned when the normalized text contains an English or Danish beef term, including *beef*, *okse*, *oksekød*, *kvæg*, *kalv*, *veal*, *entrecôte*, or *roastbeef*. The comparison group consists of classified foods not exposed as livestock treatment items: poultry; fish and seafood; eggs; fruit and vegetables; grains and bread; plant oils; sweets and snacks; beverages; plant proteins; and pantry goods. Pork, lamb, sheep, goat, dairy, and products matching more than one livestock category are never controls for beef. Prepared foods containing terms such as *pizza*, *lasagne*, *ready meal*, or *sandwich* are flagged as ambiguous when a livestock term is present. Non-food keywords are removed before commodity matching, and unmatched products are excluded rather than silently assigned to the control group.

The event month, June 2024, is excluded from both pre and post. The scraped panel begins in October 2023 and the analysis retains post-announcement months through September 2025, for 23 observed months in total. October 2025, the sixteenth full post-announcement month, and all later months are excluded. The source panel’s final relative-time value is 15; because removing June closes the event-time gap, it is displayed as event month 14. The Danish Veterinary and Food Administration records a bovine viral diarrhoea (BVD) outbreak from October 2025 to May 2026 in three cattle holdings and notes that BVD can reduce appetite and milk output (Danish Veterinary and Food Administration, 2026). A veterinary report states that surveillance detected the affected large dairy herd in September 2025 (Dansk Veterinærtidsskrift, 2025). Ending in September does not prove that all anticipatory disease effects are absent, but it excludes the officially dated outbreak interval and avoids mechanically attributing the subsequent cattle-specific supply disturbance to the tax announcement.

3.2 Official aggregate consumer prices

The second source is Statistics Denmark’s monthly PRIS01 consumer price index by COICOP 2018 product class. Beef and veal (011221) is the treated series. Candidate donors are terminal food and non-alcoholic-beverage classes with complete coverage in the estimation window. I exclude pork; lamb, sheep, and goat; dried or salted meat; offal; meat preparations; milk, yoghurt, cheese, cream, dairy desserts, other dairy products; and butter and other dairy fats. These categories are either directly livestock-exposed or sufficiently close to treated animal products that they could absorb the same announcement response. The resulting balanced panel contains beef and 52 donors.

The official-data window runs from April 2023 through September 2025: 15 months before July 2024 and 15 months from July 2024 onward. June 2024 is the final pre-treatment reference month for event-study purposes. This symmetric window prevents the aggregate estimates from being driven by unequal pre- and post-support and, like the microdata restriction, ends before the official October 2025 BVD outbreak interval.

3.3 EU robustness data

Two robustness checks use the European Commission’s Agri-food Data Portal. The first uses weekly beef-carcass prices reported by EU Member States. I retain the AO2 young-bull series, average prices in euros per 100 kg within country-month, exclude the EU aggregate, and keep the 23 Member States observed in every month from April 2023 through September 2025. Denmark is the exposed country and the other 22 Member States form the donor pool.

The second check uses the European Commission’s Beef Trade Data Explorer, which draws on Eurostat COMEXT data under statistical regime 4. The data report extra-EU beef trade by Member State, external partner, month, and CN8 product. I aggregate imports across beef products to monthly carcass-weight tonnes and balance the panel by coding unreported importer-partner-months as zero. The resulting panel contains 383 importer-partner pairs, including 17 pairs with Denmark as importer, over the same 30-month window.

Table 1 summarizes the two main estimation samples. The scraped panel contains 151,933 monthly observations from 16,912 product–store pairs, of which 5,123 observations and 592 pairs are classified as beef. Normalized prices are strongly right-skewed: the overall mean is DKK 518 per kilogram or litre, compared with a median of DKK 100, and the beef mean is DKK 266 compared with a median of DKK 127. This dispersion motivates logarithmic estimation while retaining prices in levels for transparent description. Product availability is intermittent, with the median product–store observed for eight months overall and 7.5 months for beef. The official panel is balanced by construction: 30 observations for beef and 1,560 for the 52 donor series. Its CPI distribution is much tighter, reflecting aggregation within official product classes rather than product-level price heterogeneity.

Table 1: Descriptive statistics for the estimation samples

Sample	Variable	Observations	Mean	SD	P25	Median	P75
<i>Panel A. Scraped product–store prices</i>							
All	Normalized price	151,933	518.19	4,337.95	51.84	100.00	183.33
Beef	Normalized price	5,123	265.64	1,925.50	70.00	127.00	209.75
Controls	Normalized price	146,810	527.00	4,398.05	51.18	100.00	182.14
All	Months per product–store	16,912	8.98	6.26	3	8	14
Beef	Months per product–store	592	8.65	5.79	4	7.5	13
Controls	Months per product–store	16,320	9.00	6.27	3	8	15
<i>Panel B. Statistics Denmark consumer price indices</i>							
All	Consumer price index	1,590	97.76	6.87	95.25	98.73	101.22
Beef	Consumer price index	30	89.43	7.41	84.74	86.58	89.73
Donors	Consumer price index	1,560	97.92	6.77	95.47	98.79	101.24

Source: dagligepriser.dk (Panel A); Statistics Denmark, PRIS01 (Panel B).

Notes: Scraped prices are normalized to DKK per kilogram or litre. For the months-per-product–store rows, observations count product–store units. Panel B covers the balanced April 2023–September 2025 window; its observation count is series–months. Donors exclude livestock-exposed food categories.

4 Empirical Strategy

The microdata DiD is

$$\log p_{it} = \alpha_i + \lambda_t + \beta(Beef_i \times Post_t) + \varepsilon_{it}, \quad (10)$$

where p_{it} is the normalized price of product–store pair i in month t ; α_i and λ_t are product–store and month fixed effects; $Beef_i$ identifies beef products; $Post_t$ identifies full months after the announcement; β is the average treatment effect on the treated (ATT); and ε_{it} is the error term. Standard errors are clustered by product–store. The event study replaces the post indicator with relative-month interactions and omits May 2024, the final observed pre-announcement month.

The official-data lags-only DiD preserves the earlier specification. For each month, I subtract the unweighted mean log CPI of the 52 donors from log beef CPI and regress this gap on the post indicator. Newey–West standard errors use two lags. The corresponding official event study

interacts beef with month indicators, controls for calendar month and three- and four-digit COICOP categories, and uses heteroskedasticity-robust standard errors.

Synthetic DiD follows Arkhangelsky et al. (2021), jointly weighting donor series and pre-treatment periods to construct a comparison path for beef. It operates on the balanced official panel and uses placebo inference with 200 replications.

Both main DiD designs require the retained control foods to be independent of beef with respect to the announcement: they must not be close substitutes or complements whose prices respond directly to the same news. This exclusion restriction may be strong in a general-equilibrium setting because an announcement about livestock emissions could affect several food markets at once. I assess it with two additional synthetic-DiD specifications over the same 15-month pre- and post-periods. The first treats Denmark’s log AO2 beef-car cass price as exposed and uses the other EU Member States as donors. This comparison no longer relies on other foods as controls. The second tests whether Denmark’s extra-EU beef imports changed relative to other Member State-partner pairs; its outcome is the log of one plus carcass-weight tonnes. Both specifications use synthetic unit and time weights. Country-price inference is based on 200 placebos; the import specification reports a 200-replication unit-cluster bootstrap, which is feasible because Denmark contributes multiple treated importer-partner pairs. These checks do not prove independence, but they directly examine the food-control concern and one international-market spillover that could compromise the country comparison (Appendix Figures A2 and A3).

5 Results

Table 2 reports the three beef estimates. The scraped microdata DiD implies a 4.37-log-point increase (approximately 4.47 percent), with a clustered standard error of 1.63 percentage points. The official lags-only DiD gives 6.45 log points, with a Newey–West standard error of 3.06 points. An approximate equality test, treating the estimates as independent, gives

$$z = \frac{0.0645 - 0.0437}{\sqrt{0.0306^2 + 0.0163^2}} = 0.60 \quad (p \approx 0.55).$$

Thus the official and scraped DiD estimates are statistically compatible. This agreement is encouraging for the external validity of the non-official price panel, although it cannot validate every scraped product or its level normalization.

Aggregate SDiD is positive at 0.076 but its 95 percent interval, $[-0.018, 0.170]$, includes zero. The three columns therefore support a narrower conclusion than a single preferred estimate would: the two DiD designs point to a modest positive response, while synthetic reweighting is too imprecise in this small aggregate panel to determine the magnitude precisely.

Figure 1 compares event-time patterns. In the scraped panel, the shorter pre-period is noisier but the pre-announcement coefficients are jointly insignificant ($p = 0.78$), and the post coefficients are generally positive, becoming clearly positive around event month 8. In the official panel, pre-period coefficients are small and mostly statistically indistinguishable from zero, after which

Table 2: Estimated effect of the livestock-tax announcement on beef prices

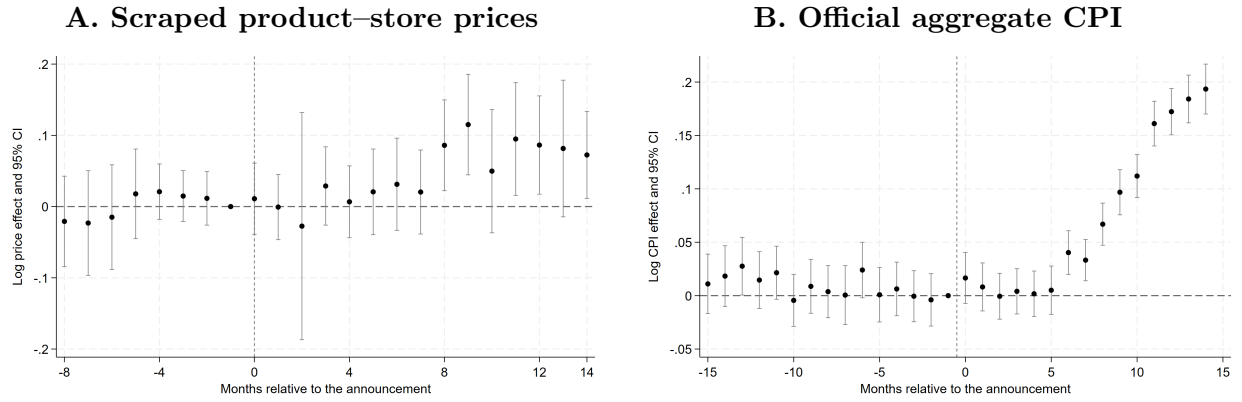
	(1)	(2)	(3)
ATT	0.0437*** (0.0163)	0.0645** (0.0306)	0.0760 (0.0481)
Observations	151,933	1,590	1,590
Units	16,912	53	53
Periods	23	30	30
Pre-treatment average	158.54	84.88	84.88
R^2	0.961	—	—
Time window	2023m10–2025m9	2023m4–2025m9	2023m4–2025m9
Lags only	No	Yes	No

Source: dagligepriser.dk (column 1); Statistics Denmark, PRIS01 (columns 2–3).

Notes: The outcome in column 1 is log normalized product price; its DiD includes product–store and month fixed effects. Parentheses contain standard errors: product–store clustered in column 1; Newey–West HAC with two lags for the unchanged lags-only aggregate DiD in column 2; and the standard error reported by Stata’s synthetic-DiD command under placebo inference with 200 replications in column 3. The observation count in column 2 is the 1,590 series–months in the official panel before cross-sectional averaging into 30 monthly gaps. The pre-treatment average is DKK/kg in column 1 and the CPI index in columns 2–3. Other livestock-exposed products are excluded from every beef control group. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

estimates rise gradually and become clearly positive around event month 6. The difference in smoothness is expected: official CPI aggregates many price quotes, whereas the scraped panel contains entry, exit, heterogeneous packaging, and intermittent product availability.

Figure 1: Beef event-study estimates across data sources



Source: dagligepriser.dk (Panel A); Statistics Denmark, PRIS01 (Panel B).

Notes: Points are relative-month coefficients and bars are 95 percent confidence intervals. In both panels, July 2024 is $t = 0$, the first complete post-announcement month. Panel A omits May 2024 ($t = -1$), because the partial event month of June is excluded, and uses product-store clustered standard errors. Panel B uses June 2024 as its omitted reference month and heteroskedasticity-robust standard errors. Both samples end in September 2025, before the official October 2025 BVD outbreak interval.

The EU country-price check yields an ATT of 0.0448 (placebo standard error 0.2043; $p = 0.826$). The estimate is not statistically distinguishable from zero, and Denmark’s series does not track its synthetic counterfactual particularly well during the pre-period (Appendix Figure A2). The point estimate is nevertheless almost identical to the scraped-product estimate of 0.0437 and close to the other main estimates. This numerical agreement is suggestive, rather than conclusive, evidence that the main findings are not generated by announcement-induced movements in the food products used as controls.

The import check yields an ATT of -0.0801 (bootstrap standard error 0.0782; $p = 0.306$), providing no evidence of an announcement-induced increase in beef imports into Denmark (Appendix Figure A3). The null result weakens the specific concern that an international supply response contaminates the country-level comparison. Taken together, the country-price and trade checks support using the main food controls as approximately independent of beef in this application, while their imprecision precludes treating that restriction as verified.

6 SCC Synthesis and Policy Calibration

The calibration uses one preferred total global SCC from each eligible paper (Cai et al., 2016; Nordhaus, 2017; Ricke et al., 2018; Cai and Lontzek, 2019; Pindyck, 2019; Hänsel et al., 2020; van den Bremer and van der Ploeg, 2021; Dietz et al., 2021; Rennert et al., 2022; Barrage and Nordhaus, 2024). Dollar values are inflated to 2024 US dollars and values per ton of carbon are converted to tons of CO₂. The 2030 value from Barrage and Nordhaus (2024) is log-interpolated between their published 2025 and 2050 path points. For the other eligible studies, a reported value at another date is treated as constant when the source does not provide a time path.

The equal-paper-weight point mean across ten studies is USD 157.96/tCO₂. The interval is constructed in two explicit layers. Index papers by j . Let θ_j be paper j ’s preferred harmonized SCC estimate and $[L_j, U_j]$ its reported positive central range. Let α_j be the probability in each tail outside that range, so the range has coverage $1 - 2\alpha_j$. Let Φ denote the standard normal cumulative distribution function and define

$$z_j = \Phi^{-1}(1 - \alpha_j), \quad \sigma_j = \frac{\log U_j - \log L_j}{2z_j}.$$

Here z_j is the corresponding standard-normal quantile and σ_j is the calibrated log-scale standard deviation. For a selected paper with a usable range, the within-paper draw is

$$X_j = \exp(\log \theta_j - \tfrac{1}{2}\sigma_j^2 + \sigma_j Z_j), \quad Z_j \sim N(0, 1),$$

where Z_j is an independent standard-normal variate, X_j is the simulated SCC value, and $E[\cdot]$ denotes expectation. The $-\sigma_j^2/2$ adjustment ensures that $E[X_j] = \theta_j$, while the range’s log width determines dispersion. For papers without a positive range, $X_j = \theta_j$. The 66 percent model range in Ricke et al. (2018) has $\alpha_j = 0.17$ and therefore $z_j = \Phi^{-1}(0.83)$; the 5th–95th quantiles in Rennert

et al. (2022) have $\alpha_j = 0.05$ and $z_j = \Phi^{-1}(0.95)$; and the ethical-parameter range in Hänsel et al. (2020) is conservatively assigned 95 percent coverage, so $\alpha_j = 0.025$ and $z_j = \Phi^{-1}(0.975)$.

In the second layer, each of 10,000 bootstrap replications samples ten papers with replacement from the ten-paper pool, makes an independent within-paper draw for every sampled paper, and records the arithmetic mean of those ten draws. Resampling papers captures between-paper dispersion; the lognormal step propagates the reported within-paper range. The 2.5th and 97.5th percentiles of the 10,000 replicated means are USD 77.22 and USD 363.40/tCO₂. Because the inputs mix model, parameter, and distributional uncertainty rather than comparable sampling errors, this interval is a hierarchical sensitivity envelope, not a frequentist confidence interval for a common structural parameter.

Figure 2 shows each preferred estimate after mapping it into the additional log beef-price increase required above the DKK 300/tCO₂e marginal abatement incentive. Filled points enter the pool. Hollow points are external comparisons: a partial mortality SCC and a synthesis that could double count the underlying literature (Carleton et al., 2022; Moore et al., 2024). The triangle is the lower bound reported by Bilal and Känzig (2026) and also remains external. The wide intervals reveal why retaining study-level variation materially changes the calibration.

Let $\bar{p} = 158.54$ be the mean pre-announcement beef price in DKK/kg, $r^{FX} = 6.8953$ DKK/USD the exchange rate, and $\chi = 59.6$ kg CO₂e/kg beef the lifecycle carbon-intensity benchmark. The benchmark follows the OECD’s presentation of the beef-herd estimate from Poore and Nemecek (2018) (OECD, 2025). It includes lifecycle emissions beyond the farm-level emissions base to which the Danish levy applies. Equation (11) is therefore a deliberately broad accounting sensitivity and can overstate the price-equivalent wedge if χ is interpreted as taxable emissions intensity. For SCC value s in USD/tCO₂ and Danish rate τ in DKK/tCO₂, define the additional log price required as

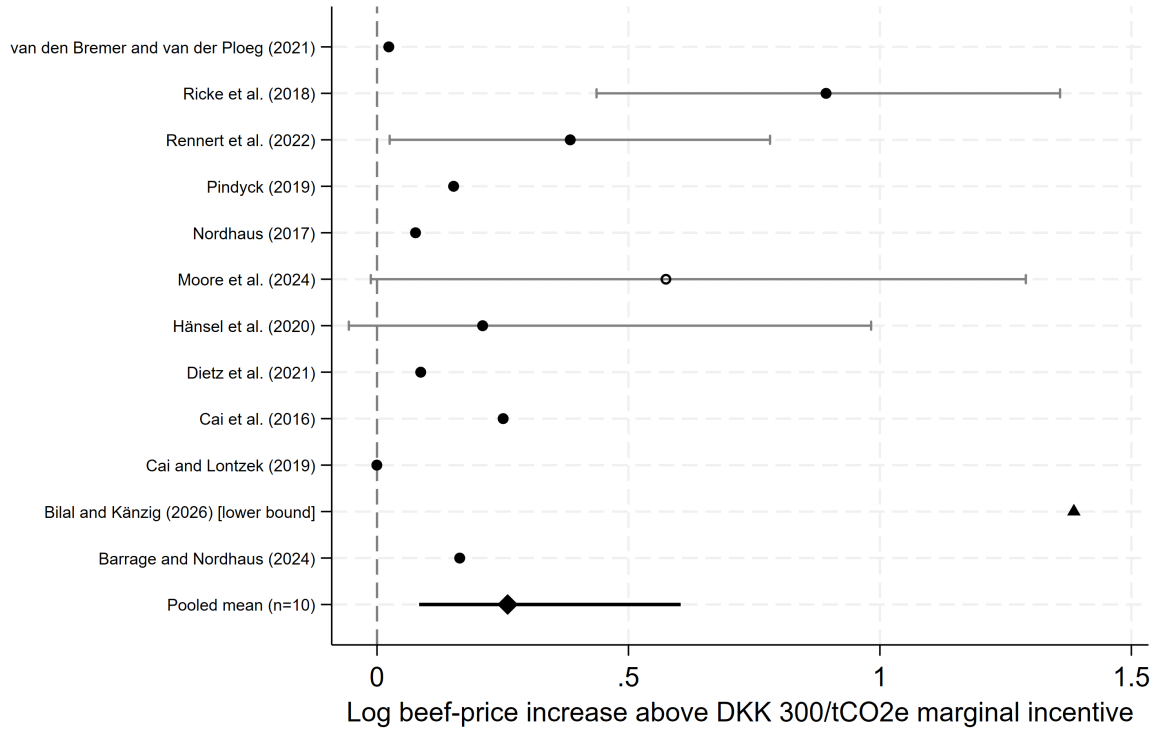
$$g(s; \tau) = \log \left[\frac{\bar{p} + (sr^{FX} - \tau)\chi/1000}{\bar{p}} \right]. \quad (11)$$

Here s denotes the SCC input and τ is evaluated separately at DKK 120 and DKK 300. The former is the effective average output burden created by the 60 percent per-animal deduction; the latter is the 2030 marginal abatement incentive (Danish Government, 2024; Danish Ministry of Taxation, 2024). At the central calibration, $s = \bar{s} \equiv J^{-1} \sum_{j=1}^J E[X_j]$, where $J = 10$ is the number of eligible papers. Because the within-paper draws are centered so that $E[X_j] = \theta_j$, $\bar{s}$ is the equal-paper-weight mean. In each bootstrap replication, s is replaced by that replication’s arithmetic mean, so the mapped interval uses the same simulated uncertainty as the SCC interval.

At the pooled mean, the SCC-implied gap is 0.311 under the DKK 120 average burden, with a pooled interval of 0.144–0.640. Under the DKK 300 marginal incentive, it is 0.260, with an interval of 0.084–0.604. Figure 3 compares both accounting benchmarks with all three econometric estimates. The micro DiD interval lies below both pooled intervals; the aggregate DiD and SDiD intervals overlap the lower tail of the DKK 300 scenario but not that of the DKK 120 scenario.

The back-of-the-envelope magnitudes are informative but are neither a welfare calculation nor an

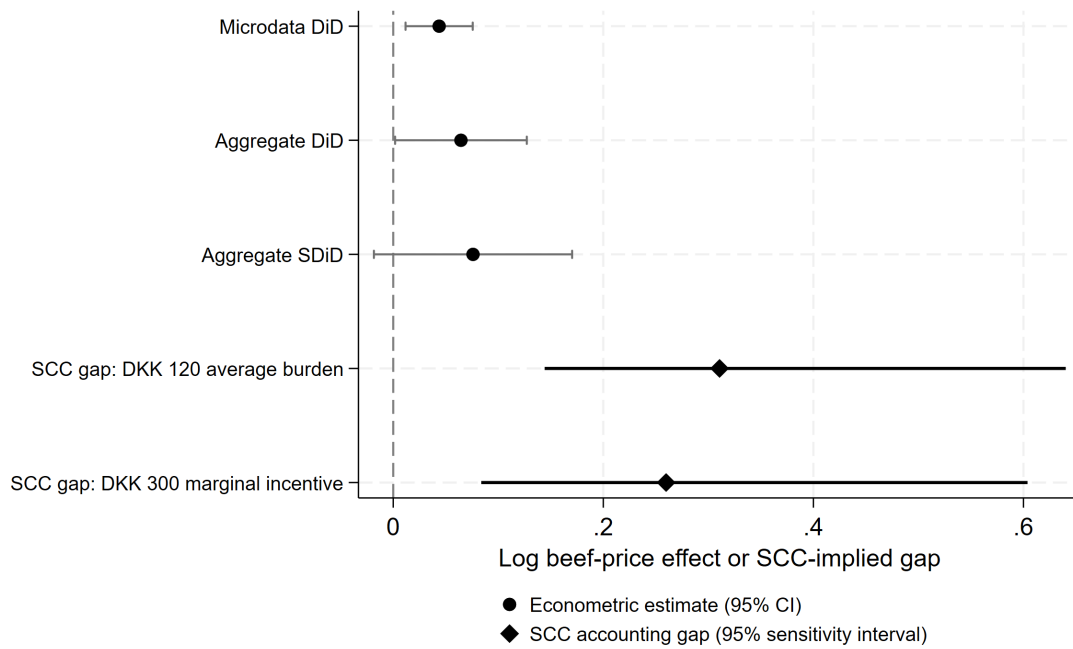
Figure 2: SCC estimates and implied beef-price gaps



Source: The studies listed in Appendix Table A1; OECD (2025).

Notes: Each point maps a preferred SCC estimate into the log beef-price increase needed above the DKK 300/tCO₂e marginal incentive, using a 59.6 kg CO₂e/kg lifecycle benchmark, 6.8953 DKK/USD, and the scraped pre-announcement beef price. The lifecycle benchmark is broader than the farm-level emissions base subject to the Danish levy, so the mapping is an accounting sensitivity rather than a tax-base calculation. Source ranges preserve their original meaning. Filled circles enter the equal-paper-weight pool; hollow circles and the lower-bound triangle do not. The pooled diamond and line show the hierarchical mean and its 95 percent interval.

Figure 3: Econometric estimates and SCC-implied gaps under two policy margins



Source: Statistics Denmark, PRIS01; dagligepriser.dk; the studies listed in Appendix Table A1.

Notes: Points show the three ATT estimates in Table 2 and the SCC-implied log gaps under the DKK 120/tCO₂e effective average output burden and DKK 300/tCO₂e marginal abatement incentive. Horizontal bars are 95 percent econometric intervals or the mapped 2.5th–97.5th percentile hierarchical SCC sensitivity interval. The 59.6 kg CO₂e/kg lifecycle benchmark is broader than the levy tax base. These are accounting comparisons, not estimates of the taxable emissions base, tax incidence, or welfare.

incidence estimate. Translating the micro DiD point estimate mechanically gives about DKK 7.1/kg at the pre-period mean, equivalent to roughly USD 17/tCO₂ under the common emissions-intensity assumption. This is an announcement-period price equivalent, not a statutory tax component: it must not be added to a future rate because the two signals occur at different dates and the model’s pass-through depends on demand curvature, adjustment costs, discounting, credibility, and contracts. The SCC exercise therefore remains an accounting sensitivity under two distinct statutory margins.

7 Discussion of the Results

Two independent DiD designs find a modest increase in beef prices following Denmark’s announcement. Their statistical compatibility is notable because the sources differ sharply in provenance, sampling, and aggregation. It suggests that carefully classified scraped prices can recover economically relevant aggregate movements faster and at much greater cross-sectional scale than official indices alone. That is evidence in favor of reliability and external validity for this application, not a blanket endorsement of non-official data: product coverage is selected, price normalization is noisy in levels, and product availability is intermittent.

Aggregate SDiD prevents an overly tidy conclusion: it is positive but imprecise. With only 53 official series and 15 pre-treatment months, synthetic weighting is intrinsically fragile. The preferred interpretation therefore rests on triangulation: the two DiD estimates agree, their event studies do not show differential pre-trends, and SDiD points in the same direction without determining the magnitude precisely.

The EU checks address a separate identification concern. Their point estimates provide no evidence that the main result is driven by movements in control-food prices or by a post-announcement increase in beef imports. The close numerical agreement between the country-price ATT and the product-level ATT is encouraging, but the country-price pre-fit is weak and both checks are imprecise. They therefore support, without establishing, the independence restriction underlying the main designs.

The meta-analytical calibration places the estimated announcement effect below the central carbon-price gap while preserving substantial uncertainty across the underlying SCC literature. The lower end of the pooled range overlaps some aggregate econometric intervals, so the evidence does not identify a single corrective tax with useful precision.

The response’s sign and timing are policy-relevant. Proposition 1 predicts a price increase before implementation when credible future taxation contracts planned supply and adjustment is costly. A demand-driven increase is less natural in this setting: news that highlights beef’s environmental cost would ordinarily weaken, not strengthen, beef demand. The agreement of two price sources, a similarly signed country-price estimate, and no detectable increase in imports therefore favor planned supply contraction. The gradual appearance of the official response around event month 6 and the scraped response around month 8 is also consistent with the contract-renewal filter in

(8). These patterns favor the model’s channel but do not directly reveal producer plans or contract dates.

The fixed-stock counterexample disciplines this interpretation. If producers merely accelerate slaughter to avoid a future tax, current supply rises and the current price falls. The observed positive price response weighs against that pure mechanism. Yet price-only data cannot determine whether quantities, herd size, production, sales volumes, or emissions actually declined. Likewise, the null import response is informative about one offsetting margin but does not measure domestic output. Establishing the full mechanism requires slaughter, herd, production, procurement-contract, and sales-volume data.

A credible announcement can thus be a useful part of policy implementation: by changing the expected future wedge, it may begin addressing the externality before collection. This is not a substitute for the tax itself. Credibility is essential, the welfare result is local, and a poorly designed policy can generate sign-reversing intertemporal substitution. Announcement and implementation should therefore be analyzed jointly rather than treating pre-collection adjustment as either zero or mechanically additive to the later statutory rate.

Three limitations are central. First, the main design identifies a national announcement relative to other food prices and therefore depends on the absence of beef-specific shocks before October 2025. The country-price and trade checks address two implications of this concern but cannot rule out every domestic or international spillover. Surveillance detected an affected dairy herd in September 2025, the final sample month, so excluding the official October outbreak interval does not rule out anticipatory cattle-market effects. Second, text rules and quantity parsing can misclassify products or create extreme normalized levels, although logs, fixed effects, explicit exclusion rules, and agreement with official CPI mitigate this concern. Third, SCC papers estimate heterogeneous objects under different discounting, damage, and uncertainty assumptions. The dual-rate calibration represents that heterogeneity and the policy’s distinct average and marginal margins; its lifecycle intensity is broader than the taxable emissions base, and it does not resolve incidence or identify the optimal Danish rate.

8 Conclusion

Denmark’s livestock-tax announcement was followed by a modest beef-price increase in two independent sources. Their close DiD estimates support carefully classified web-scraped prices for timely evaluation, while wider SDiD uncertainty cautions against a precise causal magnitude. The theoretical framework explains the positive sign as planned supply contraction under a credible future output wedge and costly adjustment; staggered contracts can explain delayed retail pass-through. A fixed-stock preemptive-slaughter mechanism predicts the opposite price sign.

The pooled SCC benchmark leaves a sizeable but uncertain accounting gap under both the DKK 120 effective average burden and the DKK 300 marginal incentive. The core policy implication is that an announcement may start addressing an externality before the tax is collected, but its

effect is not an extra statutory component and cannot be inferred from prices alone. Future work should test the mechanism with slaughter, herd, production, sales-volume, contract, realized-incidence, and farm-level emissions data, using longer panels that separate announcement effects from implementation and later supply shocks.

References

- Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., and Wager, S. (2021). Synthetic difference-in-differences. *American Economic Review*, 111(12):4088–4118.
- Barrage, L. and Nordhaus, W. D. (2024). Policies, projections, and the social cost of carbon: Results from the DICE-2023 model. *Proceedings of the National Academy of Sciences*, 121(13):e2312030121.
- Bilal, A. and Känzig, D. R. (2026). The macroeconomic impact of climate change: Global versus local temperature. *Quarterly Journal of Economics*, 141(2):889–944.
- Cai, Y., Lenton, T. M., and Lontzek, T. S. (2016). Risk of multiple interacting tipping points should encourage rapid CO₂ emission reduction. *Nature Climate Change*, 6:520–525.
- Cai, Y. and Lontzek, T. S. (2019). The social cost of carbon with economic and climate risks. *Journal of Political Economy*, 127(6):2684–2734.
- Carleton, T. et al. (2022). Valuing the global mortality consequences of climate change accounting for adaptation costs and benefits. *Quarterly Journal of Economics*, 137(4):2037–2105.
- Danish Government (2024). Regeringen og parterne i grøn trepart indgår historisk aftale om et grønt danmark. Government announcement.
- Danish Ministry of Taxation (2024). Faktaark: Skatteinitiativer i aftale om et grønt danmark. Government fact sheet.
- Danish Veterinary and Food Administration (2026). Bovin virus diarré (BVD).
- Dansk Veterinærtidsskrift (2025). Kampen mod de usynlige fjender hos kvæg.
- Dietz, S., Rising, J., Stoerk, T., and Wagner, G. (2021). Economic impacts of tipping points in the climate system. *Proceedings of the National Academy of Sciences*, 118(34):e2103081118.
- Draghi, M. (2012). Verbatim of the remarks made by mario draghi. European Central Bank.
- Fabra, N. and Reguant, M. (2014). Pass-through of emissions costs in electricity markets. *American Economic Review*, 104(9):2872–2899.
- Hänsel, M. C. et al. (2020). Climate economics support for the UN climate targets. *Nature Climate Change*, 10:781–789.
- Moore, F. C. et al. (2024). Synthesis of evidence yields high social cost of carbon due to structural model variation and uncertainties. *Proceedings of the National Academy of Sciences*, 121(52):e2410733121.

- Nordhaus, W. D. (2017). Revisiting the social cost of carbon. *Proceedings of the National Academy of Sciences*, 114(7):1518–1523.
- OECD (2025). *Measuring Carbon Footprints of Agri-food Products*. OECD Publishing, Paris.
- Pindyck, R. S. (2019). The social cost of carbon revisited. *Journal of Environmental Economics and Management*, 94:140–160.
- Poore, J. and Nemecek, T. (2018). Reducing food’s environmental impacts through producers and consumers. *Science*, 360(6392):987–992.
- Rennert, K. et al. (2022). Comprehensive evidence implies a higher social cost of CO₂. *Nature*, 610:687–692.
- Ricke, K., Drouet, L., Caldeira, K., and Tavoni, M. (2018). Country-level social cost of carbon. *Nature Climate Change*, 8:895–900.
- Rittenhouse, K. and Zaragoza-Watkins, M. (2018). Anticipation and environmental regulation. *Journal of Environmental Economics and Management*, 89:255–277.
- Rosen, S., Murphy, K. M., and Scheinkman, J. A. (1994). Cattle cycles. *Journal of Political Economy*, 102(3):468–492.
- Smulders, S., Tsur, Y., and Zemel, A. (2012). Announcing climate policy: Can a green paradox arise without scarcity? *Journal of Environmental Economics and Management*, 64(3):364–376.
- van den Bremer, T. S. and van der Ploeg, F. (2021). The risk-adjusted carbon price. *American Economic Review*, 111(9):2782–2810.

A Theory Appendix

A.1 Proof of Proposition 1

The interior first-order conditions for (3) are

$$G'_0(Q_0) + \delta\kappa(Q_T - Q_0) = 0, \quad (12)$$

$$G'_T(Q_T) - \omega - \kappa(Q_T - Q_0) = 0. \quad (13)$$

Define $h_0 \equiv -G''_0(Q_0) > 0$, $h_T \equiv -G''_T(Q_T) > 0$, and

$$\Delta \equiv (h_0 + \delta\kappa)(h_T + \kappa) - \delta\kappa^2 = h_0h_T + h_0\kappa + \delta\kappa h_T > 0.$$

Implicit differentiation of (12)–(13) gives

$$\begin{bmatrix} h_0 + \delta\kappa & -\delta\kappa \\ -\kappa & h_T + \kappa \end{bmatrix} \begin{bmatrix} \partial Q_0 / \partial \omega \\ \partial Q_T / \partial \omega \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Cramer's rule therefore yields

$$\frac{\partial Q_0}{\partial \omega} = -\frac{\delta\kappa}{\Delta} < 0, \quad \frac{\partial Q_T}{\partial \omega} = -\frac{h_0 + \delta\kappa}{\Delta} < 0. \quad (14)$$

Since $P'_0(Q_0) < 0$, $\partial P_0 / \partial \omega = P'_0(Q_0) \partial Q_0 / \partial \omega > 0$. The first expression in (14) is zero when $\kappa = 0$ or $\delta = 0$; when $\pi = 0$, $\omega = \pi\tau\chi = 0$ and there is no policy-induced change. This proves the proposition.

For completeness, substituting $G'(Q) = a - c - hQ$ into (12)–(13) and solving gives (5)–(6). Let $D \equiv h[h + \kappa(1 + \delta)]$. The coefficient $A \equiv \delta\kappa/D$ governing the announcement response satisfies

$$\frac{\partial A}{\partial \delta} = \frac{\kappa h(h + \kappa)}{D^2} > 0, \quad \frac{\partial A}{\partial \kappa} = \frac{\delta h^2}{D^2} > 0.$$

Together with $\omega = \pi\tau\chi$, these derivatives establish the comparative statics stated below (7).

A.2 Multi-period anticipation

Consider dates $t = 0, \dots, N$ and implementation at $T \leq N$. The multi-period potential is

$$\sum_{t=0}^N \beta^t [G_t(Q_t) - \omega \mathbf{1}\{t \geq T\} Q_t] - \sum_{t=1}^N \beta^t \frac{\kappa}{2} (Q_t - Q_{t-1})^2. \quad (15)$$

Let H be its Hessian in $Q = (Q_0, \dots, Q_N)'$. Strict concavity makes $A \equiv -H$ positive definite. Its diagonal entries are positive, its only off-diagonal entries are non-positive adjacent adjustment terms, and for $\kappa > 0$ it is irreducible. Hence A is a nonsingular tridiagonal M-matrix and A^{-1} is

elementwise nonnegative. Differentiating the first-order conditions with respect to ω gives

$$A \frac{\partial Q}{\partial \omega} = -b, \quad b_t = \beta^t \mathbf{1}\{t \geq T\} \geq 0.$$

Therefore $\partial Q / \partial \omega = -A^{-1}b \leq 0$, with strict inequality at every date connected by a positive adjustment link to an implementation or post-implementation date. A credible future wedge thus propagates backward through the production plan.

In the stationary linear-quadratic case, the homogeneous pre-implementation response obeys

$$\kappa q_{t-1} - [h + \kappa(1 + \beta)]q_t + \beta \kappa q_{t+1} = 0,$$

where $q_t \equiv \partial Q_t / \partial \omega$. On an infinite horizon, or locally away from the boundaries of a long finite horizon, the stable backward component is proportional to r^{T-t} , with

$$r = \frac{B - \sqrt{B^2 - 4\beta\kappa^2}}{2\kappa} \in (0, 1), \quad B \equiv h + \kappa(1 + \beta). \quad (16)$$

Thus the stable interior component of the anticipation response declines geometrically with distance from implementation. On an arbitrary finite horizon, the exact path is a linear combination of both characteristic roots, with coefficients fixed by the initial and terminal boundary conditions; equation (16) is not an exact global finite-horizon solution. The M-matrix sign result above remains exact. The geometric characterization is the multi-period counterpart to the role of $\delta = \beta^T$ in the two-date model.

A.3 Welfare and the Pigouvian benchmark

With emissions $E_t = \chi Q_t$, social welfare in the two-date model is

$$W = G_0(Q_0) - D_0(\chi Q_0) + \delta \left[G_T(Q_T) - D_T(\chi Q_T) - \frac{\kappa}{2}(Q_T - Q_0)^2 \right]. \quad (17)$$

At a given date, equating the private output wedge $\chi \tau_t$ to marginal environmental damage per unit of output $\chi D'_t(E_t)$ gives $\tau_t^P = D'_t(E_t) = SCC_t$, proving (9). This remains the first-best per-ton wedge when the adjustment cost is a private and social resource cost.

Along the private equilibrium defined by (12)–(13), differentiation of (17) and substitution of the first-order conditions yield

$$\frac{dW}{d\omega} = -\chi D'_0(E_0) \frac{\partial Q_0}{\partial \omega} + \delta [\omega - \chi D'_T(E_T)] \frac{\partial Q_T}{\partial \omega}. \quad (18)$$

Both quantity derivatives are negative by Proposition 1. At $\omega = 0$, positive marginal damage makes both terms weakly positive and the first strictly positive. More generally, $D'_0(E_0) > 0$ and $\omega \leq \chi D'_T(E_T)$ are sufficient for a positive local welfare effect. This proves Proposition 2. It is a second-best statement about a future instrument: it does not imply that omission of a current tax

is first-best.

A.4 Fixed-stock counterexample

Let a fixed stock $H > 0$ satisfy $Q_0 + Q_T = H$. Substituting $Q_T = H - Q_0$, the private objective is

$$G_0(Q_0) + \delta[G_T(H - Q_0) - \omega(H - Q_0)].$$

Its first-order condition is $G'_0(Q_0) - \delta G'_T(H - Q_0) + \delta\omega = 0$. Strict concavity gives

$$\frac{\partial Q_0}{\partial \omega} = -\frac{\delta}{G''_0(Q_0) + \delta G''_T(H - Q_0)} > 0, \quad \frac{\partial P_0}{\partial \omega} < 0.$$

A future tax therefore accelerates slaughter when total stock cannot adjust. The opposite price sign separates pure intertemporal reallocation of a fixed stock from the planned-contraction mechanism in the main model.

B Additional Results

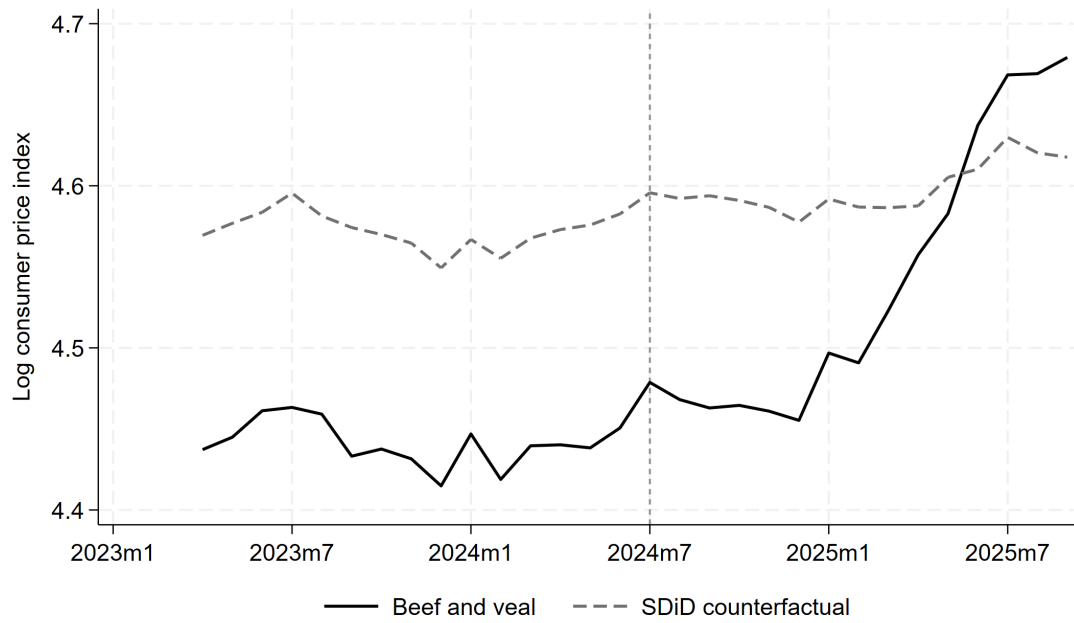
Table A1: Preferred SCC benchmarks and pooling status

Study	Method	Reference year	2024 USD/tCO ₂	Low	High	Pool status
Cai et al. (2016)	Stochastic IAM	2015	153.5	—	—	Included
Nordhaus (2017)	DICE IAM	2030	74.2	—	—	Included
Ricke et al. (2018)	Spatial IAM	2020	599.9	254.6	1,158.0	Included
Cai–Lontzek (2019)	Stochastic IAM	Undated	43.4	—	—	Included
Pindyck (2019)	Expert elicitation	Undated	107.0	—	—	Included
Hänsel et al. (2020)	Updated DICE	2020	133.7	22.5	688.2	Included
van den Bremer–van der Ploeg (2021)	Asset-pricing IAM	2015	52.7	—	—	Included
Dietz et al. (2021)	Meta-analytic IAM	2020	78.5	—	—	Included
Rennert et al. (2022)	Modular IAM	2020	224.2	53.3	500.6	Included
Barrage–Nordhaus (2024)	DICE-2023	2030	112.5	—	—	Included
Carleton et al. (2022)	Mortality damages	2020	21.0	−30.3	65.8	Partial; external
Moore et al. (2024)	Literature synthesis	2020	343.0	38.8	1,059.3	Double-count risk; external
Bilal–Känzig (2026)	Macro empirical	2024	> 1,200	—	—	Lower bound; external

Source: The studies listed in the first column.

Notes: Values are converted to 2024 US dollars per ton of CO₂. One preferred global total-SCC estimate enters per eligible paper. Values at dates other than 2030 are treated as time-invariant when no within-paper path is reported. Barrage–Nordhaus is log-interpolated to 2030. Low and high retain their source-specific interpretation. The pooled mean gives equal weight to each included paper; partial estimates, a literature synthesis, and a lower bound are external comparisons.

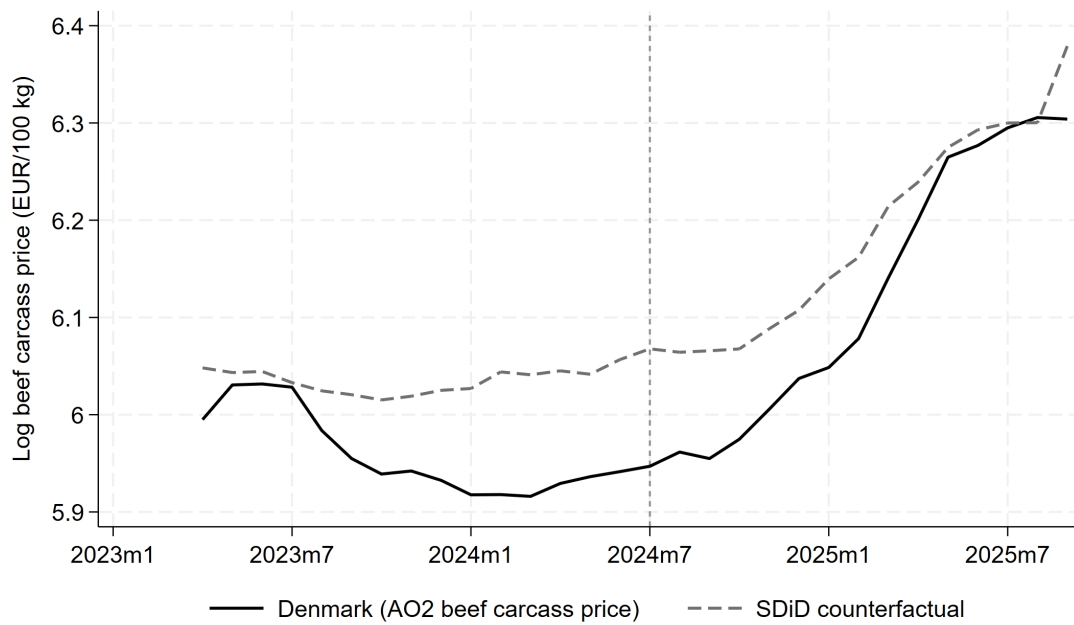
Figure A1: Official-data synthetic DiD: beef and SDiD counterfactual



Source: Statistics Denmark, PRIS01.

Notes: Series are log CPI for beef and veal and the Stata SDiD counterfactual over the symmetric 15-month pre/post window. The dashed vertical line marks July 2024, the first post-announcement month.

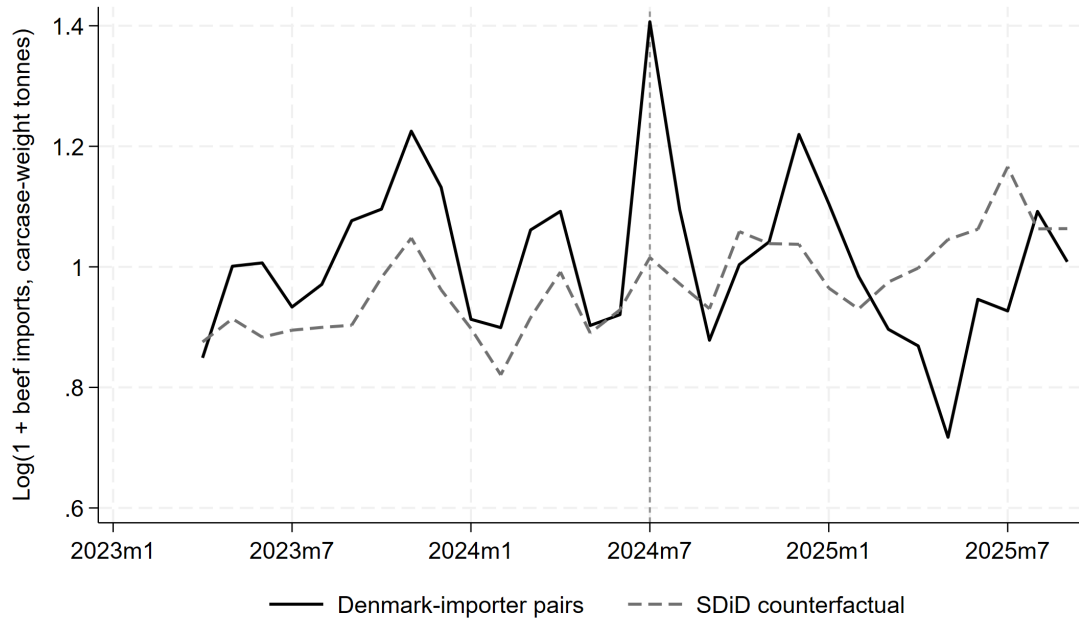
Figure A2: Country-level synthetic DiD: Danish beef-carcass prices



Source: European Commission, Agri-food Data Portal, Beef Prices.

Notes: Series are log AO2 young-bull carcass prices for Denmark and the SDiD counterfactual constructed from 22 other EU Member States over the symmetric April 2023–September 2025 window. The dashed vertical line marks July 2024. The pre-treatment fit is imperfect. The ATT is 0.0448 (placebo standard error 0.2043; $p = 0.826$; 200 replications).

Figure A3: Synthetic DiD: beef imports into Denmark



Source: European Commission, Beef Trade Data Explorer; Eurostat COMEXT.

Notes: Series are mean log-one-plus monthly extra-EU beef imports, measured in carcass-weight tonnes, for Denmark-partner pairs and their SDiD counterfactual over the symmetric April 2023–September 2025 window. The dashed vertical line marks July 2024. The ATT is -0.0801 (unit-cluster bootstrap standard error 0.0782 ; $p = 0.306$; 200 replications).